PHYTO-GENIOUS

Product Sheet

NMN

PHYTO-GENIOUS

NMN SERIES

PRODUCT INFORMATION

NMN 30000 60 Capsules

NPN: 80122085

Ingredients: 500mg Nicotinamide Mononucleotide (NMN)

Recommended Use: Helps in energy metabolism and tissue formation. Helps to maintain the body's ability to metabolize nutrients. Helps to prevent vitamin B3 deficiency. Helps in normal growth and development.

Suggested Use: (Adults) Take 1 capsule per day.

PRODUCT INFORMATION

NMN 15000

60 Capsules

NPN: 80123421

Ingredients: 250mg Nicotinamide Mononucleotide (NMN), 10mg Nicotinamide Riboside (NR), 10mg Thiamine

Recommended Use: Helps to maintain/support the body's ability to metabolize nutrients. Supports energy production. A factor in normal growth and development. Contributes to maintaining general health. Helps to prevent thiamine deficiency. Source of vitamins, factors in normal growth and development.

Suggested Use: (Adults 18 years and older) Take 2 capsules, 1 time per day. Take with food.

PRODUCT INFORMATION

NMN 10000

60 Capsules

NPN: 80123154

Ingredients: 167mg Nicotinamide Mononucleotide (NMN), 10mg Thiamine

Recommended Use: Contributes to maintaining general health. Helps to maintain the body's ability to metabolize nutrients. Supports energy production. Helps to prevent thiamine deficiency which helps support normal growth. Source of vitamin(s)/mineral(s)/vitamin(s) and mineral(s), a factor/factors in normal growth and development.

Suggested Use: (Adults 18 years and older) Take 2 capsules, 1 time per day. Take with food.

PRODUCT INFORMATION

NMN 6000

60 Capsules

NPN: 80123452

Ingredients: 100mg Nicotinamide Mononucleotide (NMN), 20mg Thiamine, 20g Niacin

Recommended Use: Helps in energy metabolism and tissue formation. Supports energy production. Helps normal growth and development. Helps to maintain/support the body's ability to metabolize nutrients. Helps to prevent vitamin B3 deficiency. Contributes to maintaining general health.

Suggested Use: (Adults 18 years and older) Take 2 capsules, 1 time per day. Take with food.

PRODUCT INFORMATION

NMN 3000

60 Capsules

NPN: 801231455

Ingredients: 50mg Nicotinamide Mononucleotide (NMN), 10mg Thiamine, 10mg Niacin

Recommended Use: Helps in energy metabolism and tissue formation. Supports energy production. Helps normal growth and development. Helps to maintain/support the body's ability to metabolize nutrients. Helps to prevent vitamin B3 deficiency. Contributes to maintaining general health.

Suggested Use: (Adults 18 years and older) Take 2 capsules, 1 time per day. Take with food.

HOW TO CHOOSE NMN?

NMN 30000

Maintain a strong body and protect cardiovascular health. Promote DNA repair, improve cognitive function and memory.

NMN 15000

Soothes the hangover and protects the liver, so you can sleep peacefully at night. Keep skin in good condition and improve physical and mental health.

NMN 10000

Maintain brain vitality at all times, improve body function, promote metabolism, awaken youthful vitality.

NMN 6000

Improve cell vitality, maintain emotional stability, relieve the pressure of life and work, and maintain a good physi-cal and mental state.

NMN 3000

Simple, quick and burden-free, keep youthful and energetic, and keep the best moments. Maintain a healthy metabolic rate.

MALE SUPPLEMENT

PRODUCT INFORMATION

VIRILITY

60 Capsules

NPN: 80121806

Ingredients: L-Arginine, Royal jelly, Maca, Ying yang huo, Saw palmetto, Damiana, Thiamine, Folate, Niacin, Tribulus.

Recommended Use: Helps to support emotional aspects of sexual health. Contributes to maintaining general health. Helps in energy metabolism and tissue formation. Helps to form red blood cells. Helps to support the body's ability to metabolize nutrients. Supports energy production. Tribulus is traditionally used in Ayurveda as Vrsya (aphrodisiac) for men.

Suggested Use: (Adults 18 years and over) 2 capsules, 1 time per day. Take with food to minimize gastric disturbance.

WIDE Naturals Inc.

WIDE Naturals Inc. is an enterprise focusing on the research and development, production and sales of health products. Our company has not only obtained NHP site license issued by Health Canada, but also registered with FDA as a licensed manufacturing facility.

WIDE Naturals has an independent product research and development team, an advanced biotech Laboratory, and a GMP standard health product manufacturing site.

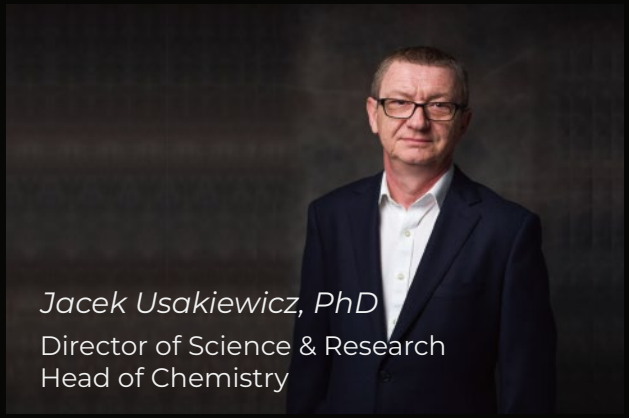
From raw ingredients to finished products, we oversee every step to guarantee consistency and strict adherence to standards.

INNOVATE BIOTECH LABORATORY

Innovate Biotech Laboratory, equipped with state-of-the-art equipment, and a R&D team with over 20 years of experience in genetics, extraction, drug development and product formulation.

Innovate Laboratories ensures that we adhere to strict standards of scientific integrity to bring highest quality Natural Health Products to our consumers.

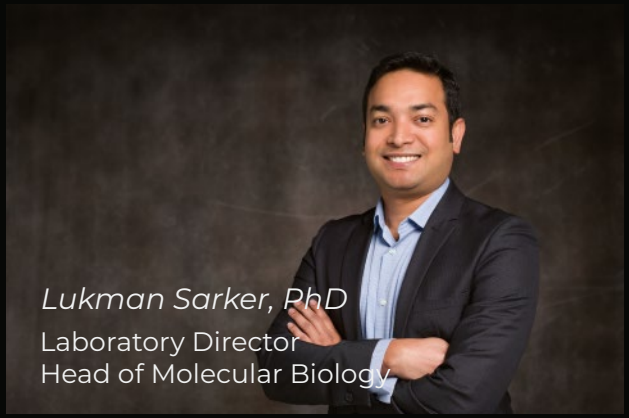
RESEARCH & DEVELOPMENT TEAM



Jacek Usakiewicz, PhD
Director of Science & Research
Head of Chemistry

Belal Hossain, Msc
Lead Microbiologist

Ayelign Adal , PhD
Molecular Biologist



Lukman Sarker, PhD
Laboratory Director
Head of Molecular Biology

Jessika Bedard-St-Amant
Analytical Chemist

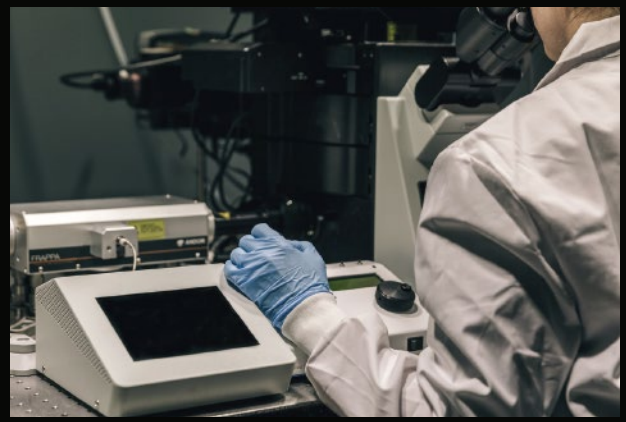
Drake Poirier
Quality Assurance &
Regulatory Affairs Manager

6000

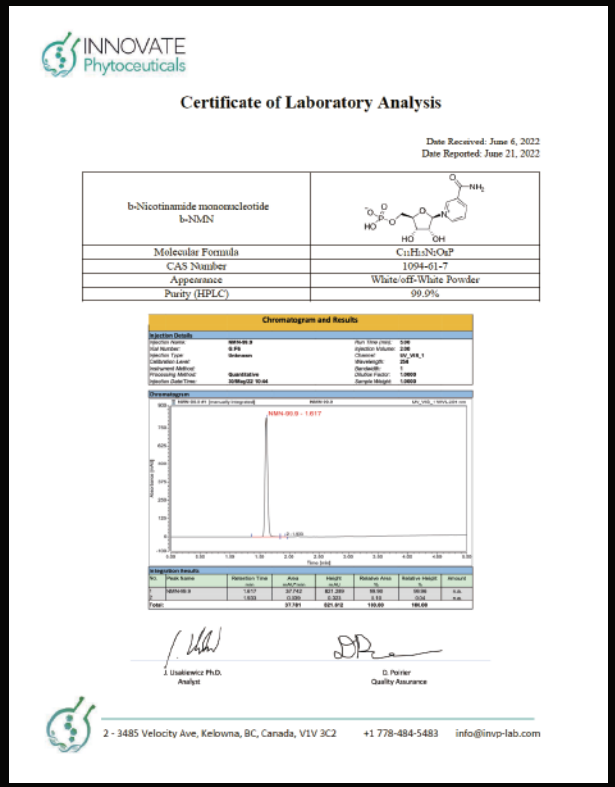
15000
THE KING
NATHANIEL WILLIAMS

10000

INDUSTRY-LEADING EQUIPMENT



NMN PURITY TEST



INNOVATE Phytoceuticals

Certificate of Laboratory Analysis

Date Received: June 6, 2022
Date Reported: June 21, 2022

b-Nicotinamide mononucleotide b-NMN	
Molecular Formula	$C_{11}H_{15}N_2O_8P$
CAS Number	1094-61-7
Appearance	White/off-White Powder
Purity (HPLC)	99.9%

Chromatogram and Results

Injection Details

Injection Name:	NMN-00.3	Run Time (min):	5.00
Vial Number:	0.F6	Injection Volume:	2.00
Injection Type:	Unknown	Channel:	UV_VIS_1
Calibration Level:		Wavelength:	254
Instrument Method:		Bandwidth:	1
Processing Method:	Quantitative	Dilution Factor:	1.0000
Injection Date/Time:	30/May/22 10:44	Sample Weight:	1.0000

Chromatogram

Integration Results

No.	Peak Name	Retention Time min	Area mAU*min	Height mAU	Relative Area %	Relative Height %	Amount
1	NMN-00.3	1.617	37.742	821.289	99.50	99.06	n.a.
2		1.833	0.189	8.383	0.50	0.94	n.a.
Total:			37.931	829.672	100.00	100.00	

J. Usakiewicz Ph.D.
Analyst

D. Poirier
Quality Assurance

2 - 3485 Velocity Ave, Kelowna, BC, Canada, V1V 3C2 +1 778-484-5483 info@invp-lab.com

NMN Purity of **99.9%**

Enzymatic Extraction Technology
Free of impurities and heavy metals.

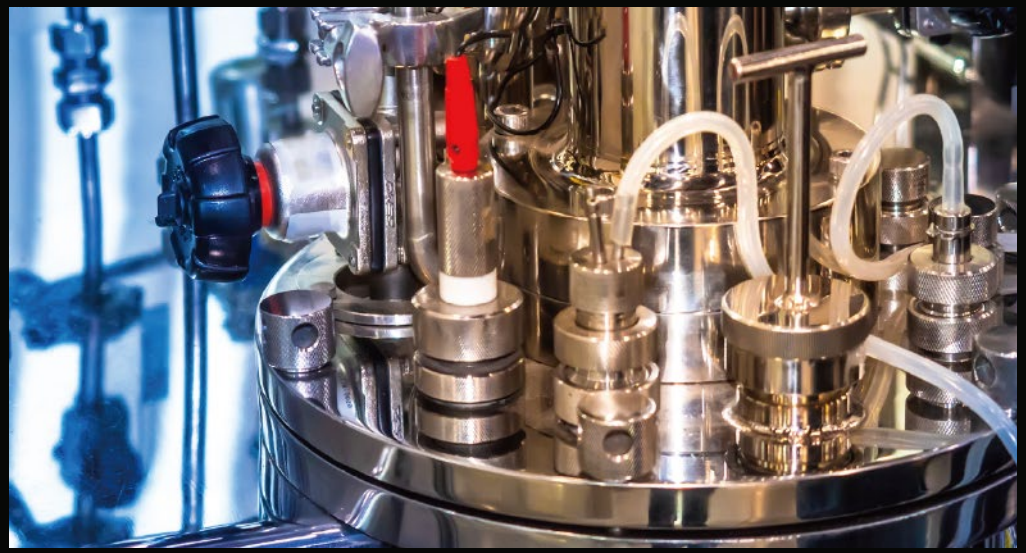
ENZYMATIC EXTRACTION TECHNOLOGY

Our NMN is obtained by enzyme extraction method with Nobel Prize-winning enzyme directed evolution technology.

By simulating the metabolic process of NMN in human body, we realized the fermentation process of *in vitro* enzymes through *in vitro* expression of in vivo enzymes.

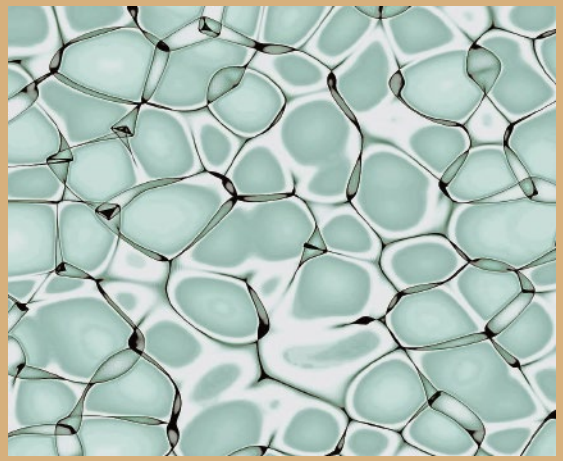
Compared with the chemical synthesis method with low costs and impurities, enzymatic extraction is safer, more stable, has the highest purity and activity, and the best conversion rate.

With a purity of 99.9%, our NMN is top quality and easier to absorb.



Enzyme extraction method simulates the physiological environment of the human body to the greatest extent, and the natural fermentation production process ensures the absorption, bioavailability, and conversion rate of the NMN.

OUR TECHNICAL ADVANTAGES



ENZYMATIC EXTRACTION

More efficient and safer. NMN is extracted by bio-enzyme preparation technology, and after a bio-enzyme-catalyzed reaction, it can achieve higher purity, higher conversion in the body and is easily absorbed by the human body, improving the effect.

VS

ORDINARY FERMENTATION

Costly, lower yield. Currently there is one brand using ordinary fermentation in Japan, yet the process is complicated and laborious, resulting in lower concentration and purity.

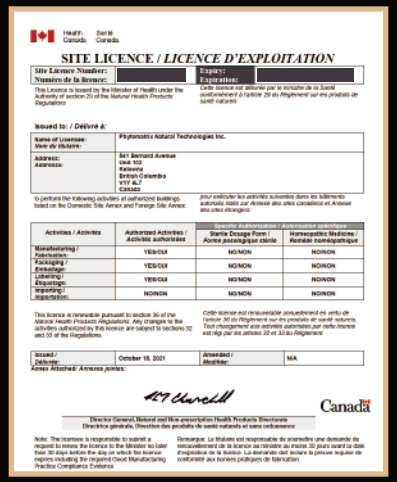
CHEMICAL SYNTHESIS

The chemical synthesis method mainly completes the synthesis of NMN through the reaction between compounds. This extraction method has simple process, high yield and very low production cost. However, due to serious chemical residues, low purity and high impurities, long-term use may also have adverse effects on the body.

QUALIFICATION

NPN

Canadian Natural Health Products (NHP) Licensed Manufacturer



GMP Standard

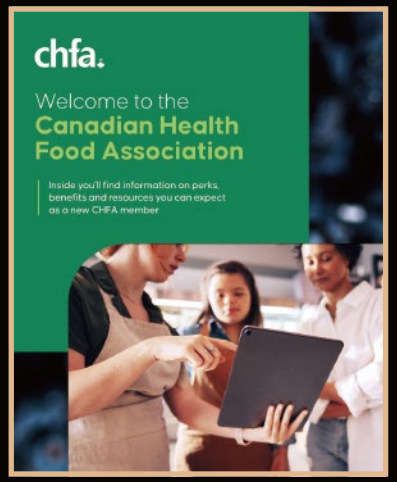
Certified by Health Canada Compliant with GMP requirements

FDA Registered

Site Licensed by FDA Compliant with FDA standard

Free Sale

FHCP attests to the product being compliant and freely sold with Canadian laws and regulations



Member of CHFA

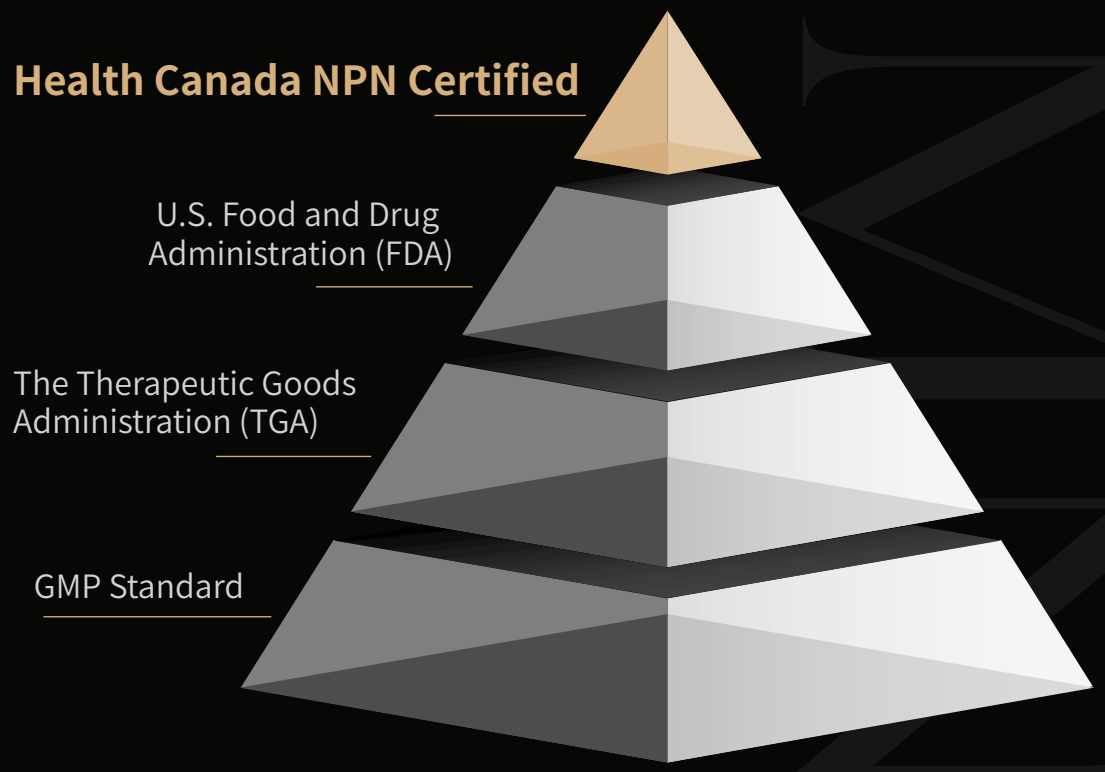
Canadian Health Food Association members are actively involved in building industry development

NPN CERTIFIED BY HEALTH CANADA

Canada's strict review of product ingredients, formulas, and efficacy is renowned and being considered as one of the most stringent drug management systems over the world.

The NPN licensed products must be assessed by Health Canada and found to be safe, effective and of high quality under their recommended conditions of use.

PHYTO-GENIOUS NMN has been reviewed by the Health Canada for more than 1 year, which is the only brand in Canada that has been approved to use 4 NMN-related advanced health claims.



ADVANTAGES

High Purity

Enzymatic Extraction Technology
NMN Purity of 99.9%
High Conversion and Absorption

Imported Ingredients

Ingredients Sourced from US
Nobel-Prize-Winning
Directed Evolution Technology
Safer and More Stable

Canadian Quality

Health Canada Assessed with
Natural Product Number
GMP Manufacturing Standard

Compound Formula

Designed for Different Age Groups
Added NR, Thiamine, Niacin

Lab Formulated

Innovate Laboratory involved in
Formulation and Testing

WHY PHYTO-GENIOUS

US BRAND

UNKOWN PURITY

Only 1 Product

- Unknown Absorption Time
- Unknown Biological Activity
- Ordinary Fermentation
- cGMP
- Not Specified R&D Conditions

JP BRAND

UNKOWN PURITY

Only 1 Product

- Unknown Absorption Time
- Unknown Biological Activity
- Unknown Extraction Method
- GMP & FDA
- Third-party Processing And Production

JP BRAND

UNKOWN PURITY

Only 2 Products

- Unknown Absorption Time
- Unknown Biological Activity
- Ordinary Fermentation
- No Qualifications
- Not Specified R&D Conditions

HK BRAND

99.89%

Only 2 Products

- High Absorption Rate
- High Biological Activity
- Ordinary Fermentation
- FDA
- Not Specified R&D Conditions

BEST SELLERS

PHYTO-GENIOUS

99.9%

PURITY

30000/15000/10000/6000/3000
5 NMN PRODUCTS

- 10 Minutes Fast Absorption
- Higher Biological Activity
- Bio-fermentation Technology
- NPN Health Canada
  NHP Manufacturer
  GMP Standard
  FDA Certified
- Innovate Biotech
  Laboratory R&D

BENEFITS OF NAD^+

NMN
PHYTO-GENIOUS

- Helps the liver metabolize alcohol
- Energizes Body
- Promotes Cellular Repair
- Increases Metabolism
- Helps Maintain Cardiovascular Health
- Promotes Better Sleep
- Improves Mood
- Improve Cognitive Concentration

WHAT DIFFERENCES CAN NMN MAKE?

Slows down body aging

Five NMN products have been selected to provide comprehensive care for individuals with different health concerns. These products can replenish the loss of NAD+ as the body ages and are scientifically formulated to achieve healthy aging.

Improves cognitive function

NMN can activate brain power and improve cognitive function. It can also improve brain memory, relieve anxiety, protects liver, and enhance the quality of sleep.

Protects overall health

NMN can promote DNA self-repair and protect the body‘s cardiovascular health.

Promotes energy metabolism

NMN promotes metabolism and can relieve the fatigue caused by busy urban lifestyles. It also helps to reduce weight gain associated with aging and slower metabolism, stimulates cell production capacity, and boosts vitality in the body.

NMN USER REVIEW

Male, 33, Banker

NMN
PHYTO-GENIOUS

3 Days

Increase water intake by 30%

Lifestyle gets more regular

Feel sleepy at 10pm

Wake up naturally at 5am

Clear mind and energetic

Endurance increased by 15%

Power increased by 15%

Metabolism increased by 15%

1 Month

More water intake

More power and endurance

Smooth detoxification

Wake up full of energy

Clear mind all-day long

Feel sleepy at 11pm

7hrs sleep daily

1.5 Months

Increased basal metabolism

Increased appetite

5% weight loss

Drink plenty of water

Easy to fall into deep sleep

Feel the body's natural repair

3 Months

Drink water frequently

Excellent sleep quality

Feel energetic everyday

Basal metabolism increased to 200 calories

NMN IN FOOD

NMN can be naturally synthesized in cells and can also be found in a variety of foods, including broccoli, cabbage, cucumber, edamame, avocado, etc., but the quantity is minuscule. Therefore, additional supplements are necessary to effectively elevate NAD+ levels.

52kg of Veggies

- 10kg Avocado
- 15kg Broccoli
- 18kg Cabbage
- 9kg Cucumber

=

1 Cap

NMN

*Data based on NMN10000.

HIGH PURITY

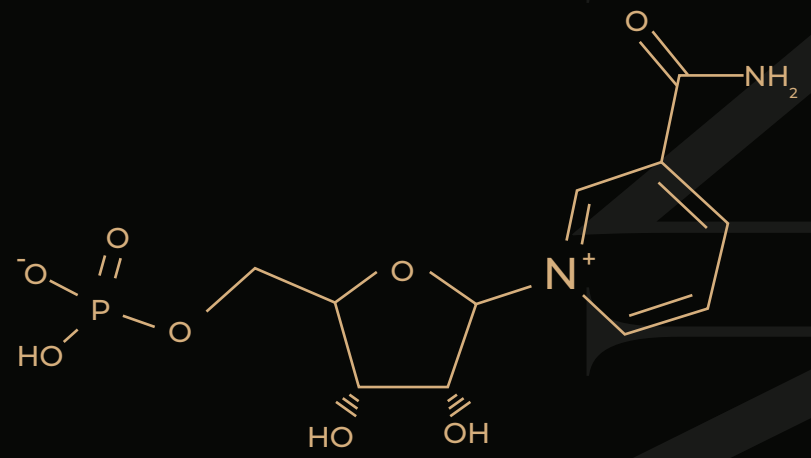
NMN
PHYTO-GENIOUS

Since NAD+ levels begin to decline as people age, NMN is intended for healthy men and women between the ages of 30 and 60 who are health-conscious. It would be most effective when used in conjunction with a healthy diet, exercise regimen, and other good habits.

Dr. Sinclair believes that only small intakes of NMN are incapable of achieving the desired anti-aging effects. Therefore, NMN users can only truly enjoy the effects of NMN by using products that are as pure as possible.

RE-EVALUATING AGING

The aging of the body was once considered an unavoidable stage in life, but in recent years, the scientific community has launched a "longevity revolution" and put forward the view that "aging is a treatable disease".



In 2015, a group of international researchers declared that aging should be categorized as a disease

In 2018, the World Health Organization (WHO) implemented an extension code for “aging related disease”

Harvard University Professor David Sinclair believes that reverse aging is within the realm of possibility in the future. Furthermore, he has also discovered many incredible age combatting capabilities of NMN at Harvard Medical School

DEFINING AGING

Common Symptoms of Aging:

Tired Easily Body Aches Insomnia

The following illnesses are related to aging:

Diabetes Stroke Cancer Heart Attack

The organs along with the immune system begins to degenerate and deteriorate with age, which also increasing the likelihood of contracting diseases and illnesses.

NMN
PHYTO-GENIOUS

WHAT IS NAD?

NAD+ is an essential coenzyme required for life and cellular function. Enzymes are the catalysts that make biochemical reactions possible. NAD+ is the most abundant molecule in the human body after water, used by many proteins in the body (such as Sirtuins) that repair damaged DNA and help generate chemical energy that the body uses.

Since its discovery in 1906, NAD+ has attracted the attention of scientists for its abundance in the human body and its key role in the molecular pathways that keep the body functioning. In animal studies, increasing the levels of NAD+ in the body has yielded encouraging results in research areas such as metabolism and age-related diseases, and has even shown some anti-aging and immune system boosting properties. The likelihood of contracting age-related diseases such as diabetes, cardiovascular and neurodegenerative diseases saw a general decline.

Therefore, “How to Replenish NAD+” became the main focus in combatting aging.

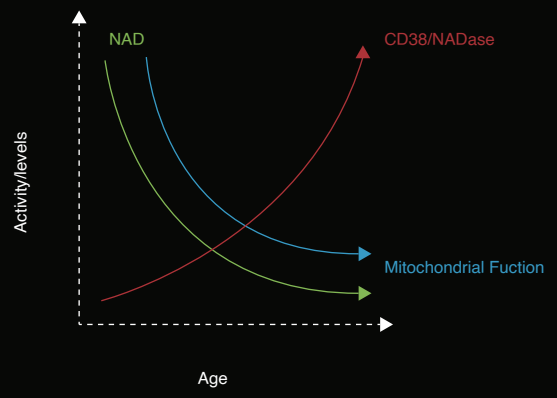
NMN
PHYTO-GENIOUS

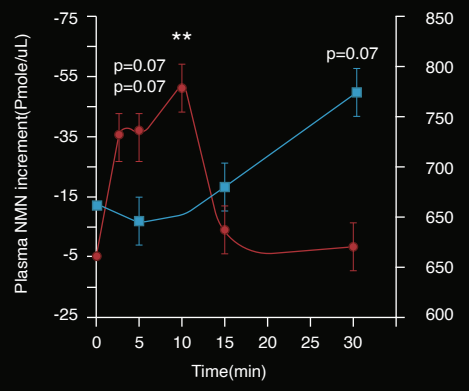
WHAT IS NMN?

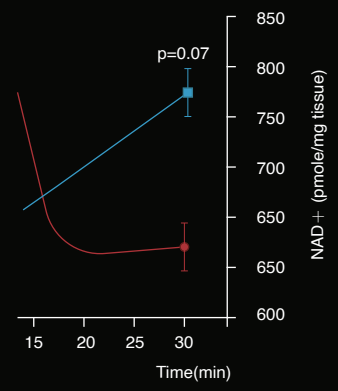
NMN (Nicotinamide mononucleotide), the full name of "β-nicotinamide mononucleotide", is a naturally occurring biologically active nucleotide. Since nicotinamide belongs to vitamin B3, NMN is a vitamin B family derivative, which is widely used in various biochemical reactions of the human body and is greatly responsible for immunity and metabolism.

NMN is the immediate precursor of NAD+ in the human body; its function is to be converted to NAD+. NAD+ (nicotinamide eliminates purine dinucleo- tides), also known as coenzyme, is the oxidized form of NAD (nicotinamide adenine dinucleotide) and exists in every cell and participates in thousands of reactions. In a variety of cellular metabolic reactions, NAD molecules play an important role and are essential for cell viability.

WHY NMN?







NAD+ Level Declines with Age

NAD+ is a key factor in youthful vitality. As we age, NAD+ decreases, bodily functions begins failing, and metabolism decreases.

Oral Intake NMN

Since the NAD+ molecule is too large for the human body to absorb directly, taking NMN as a precursor of NAD+ orally is the most effective way to increase the level of NAD+ in the body.

Raise NAD+ Level

Enough NAD+ in the body can help the body maintain cellular activity and increase metabolism.

NMN EXPERIMENT RESULTS

NAD+ DECLINES WITH AGE

Harvard Medical School experiment on mice discovered that continuous administration of NMN for 7 days was dramatic improvement in muscle function and reversed detrimental age-associated changes in muscles, and they were able to perform on average 2x the amount of activity than others of their age.

Gomes, Ana P., et al. “Declining NAD+ Induces a Pseudohypoxic State Disrupting Nuclear-Mitochondrial Communication during Aging.” Cell, vol. 155, no. 7, 19 Dec. 2013, pp. 1624–1638., doi:10.1016/j.cell.2013.11.037.

IMPROVED BRAIN ACTIVITY

Test subjects treated with NMN showed improved memory, cognitive function as well as decrease in Alzheimer’s-disease-associated pathological characteristics in University of Texas Medical School experiment.

Yao, Zhiwen, et al. “Nicotinamide Mononucleotide Inhibits JNK Activation to Reverse Alzheimer Disease.” Neuroscience Letters, vol. 647, 24 Apr. 2017, pp. 133–140., doi:10.1016/j.neulet.2017.03.027.

Poddar, Saikat Kumar, et al.“Nicotinamide Mononucleotide: Exploration of Diverse Therapeutic Applications of a Potential Molecule.”Biomolecules, vol. 9, no. 1, 2019, p. 34., doi:10.3390/biom9010034.

GENERAL WELLNESS

Tsinghua Medical School experiment discovered that NMN is shown to assist stem cell reproduction, reduce muscle inflammation and assist with telomere maintenance.

Yi, Meiqi, et al. “Comparative Proteomic Analysis Identifies Biomarkers for Renal Aging.” Aging, vol. 12, no. 21, 6 Nov. 2020, pp. 21890–21903., doi:10.18632/aging.104007.

Brett, Jamie O., et al. “Exercise Rejuvenates Quiescent Skeletal Muscle Stem Cells in Old Mice through Restoration of Cyclin D1.” Nature Metabolism, vol. 2, no. 4, 13 Apr. 2020, pp. 307–317., doi:10.1038/s42255-020-0190-0.

LIVER PROTECTION

A paper published in Human Genomics shows that NMN can work to treat liver inflammation and dysregulation of metabolism caused by consumption of alcohol. The study showed that administration of NMN can limit the damage on liver caused by alcohol.

Assiri, Mohammed A., et al. “Investigating RNA Expression Profiles Altered by Nicotinamide Mononucleotide Therapy in a Chronic Model of Alcoholic Liver Disease.” Human Genomics, vol. 13, no. 1, 10 Dec. 2019, doi:10.1186/s40246-019-0251-1.

INCREASED INSULIN SENSITIVITY

Japan Keio Medical School experiment discovered that subjects given NMN displayed increased insulin sensitivity, decreased blood sugar and overall improvements in metabolic health markers.

Yoshino, Mihoko, et al. “Nicotinamide Mononucleotide Increases Muscle Insulin Sensitivity in Prediabetic Women.” Science, vol. 372, no. 6547, 2021, pp. 1224–1229.,doi:10.1126/science.abe9985.

CARDIOVASCULAR HEALTH

Treating older test subjects with NMN significantly improved arterial health and reduced arterial stiffness compared to untreated subjects, significantly decreases risk to age-related heart diseases in experiment by University of Colorado.

Picciotto, Natalie E., et al. “Nicotinamide Mononucleotide Supplementation Reverses Vascular Dysfunction and Oxidative Stress with Aging in Mice.” Aging Cell, vol. 15, no. 3, 2016,
pp. 522–530., doi:10.1111/acel.12461.

2014 Times Magazine
Most Influential 100 people in the World

Dr. David Sinclair

Dr. David Sinclair, professor of genetics and co-director of the Paul F. Glenn Center for Biology of Aging Research at Harvard Medical School, uses 1 g of NMN everyday as an anti-aging supplement. In an interview, Dr. Sinclair claims, “if we give 1g [of NMN] to a human subject for 10 days, they will double their NAD+ levels in their blood cells, and if you give 2g, you triple it” and declares there has been no negative side effects observed in human clinical trials.

Dr. Sinclair is known for his works on understanding the process of aging and how to reverse its effects. His research primarily focuses on aging with epigenetics, having worked with other anti-aging chemical compounds such as resveratrol.

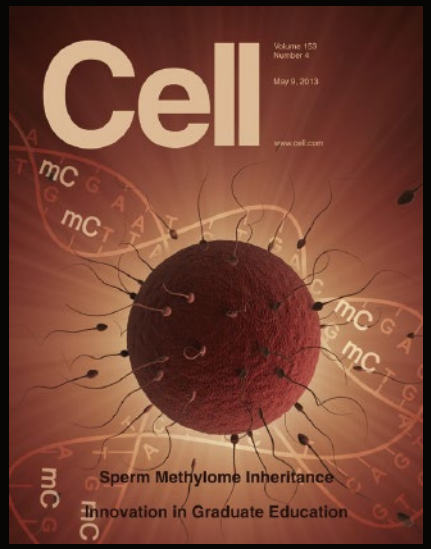
Due to his achievements in the field of aging research, in 2014, he was selected in Times Magazine as one of the most influential 100 people in the world. In 2018 Cell Metabolism paper, Dr. Sinclair concluded many therapeutic potentials of NAD+ boosting, and that when orally taken, NMN can convert to NAD+ within 15 minutes, raising NAD+ balance in the bloodstream.

Awards

Medical Research Medal
Advance Award in Life Sciences
Irving S. Wright Award of Distinction
Officer of Order of Australia

Australian Society for Medical Research (2014)
Australian Government (2017)
American Federation for Aging Research (2018)
Australian Government (2018)

SCIENCE MAGAZINES PUBLISH PAPER ON NMN



US Cell Magazine

US Science Magazine

US Nature Magazine

Research papers on NMN have been published in scientific journals. The scientific community is excited about the anti-aging effect of NMN and its great impact on the future research on the aging process

NOBEL PRIZE LAUREATES INVESTED IN THE FUTURE OF NMN

Nobel

Sir Richard Roberts
Nobel Prize in Physiology or Medicine 1993

Eric Kandel
Nobel Prize in Physiology or Medicine 2000

Daniel Kahneman
Nobel Prize in Physiology or Medicine 2002

Aaron Ciechanover
Nobel Prize in Physiology or Medicine 2004

Jack Szostak
Nobel Prize in Physiology or Medicine 2009

Martin Karplus
Nobel Prize in Physiology or Medicine 2013

Thomas Sudhof
Nobel Prize in Physiology or Medicine 2013

Paul Modrich
Nobel Prize in Physiology or Medicine 2015

These 8 Nobel Prize laureates have combined forces to oversee NMN related product research and clinical trials, showing great interests in the future of NMN as the revolutionary anti-aging supplement.

"Even if you exercise and have the heathiest diet, you're still going to have lower NAD^+ levels by the time you're in the latter half of your life."

— Dr. David Sinclair

WWW.PHYTOGENIOUS.COM

PHYTO-GENIOUS

STAY AT YOUR PEAK

CONTACT

WIDE Naturals Inc.
#1050-1040 W Georgia St
Vancouver, BC V6E 4H1
778-606-2545
info@widenaturals.com